## Question 2

Open hj.cc ~/ns-allinone-3.40/ns-3.40/scratch

ques4.cc hj.cc

```
#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/applications-module.h"
using namespace ns3;
NS_LOG_COMPONENT_DEFINE("FirstScriptExample");
int main(int argc, char *argv[]) {

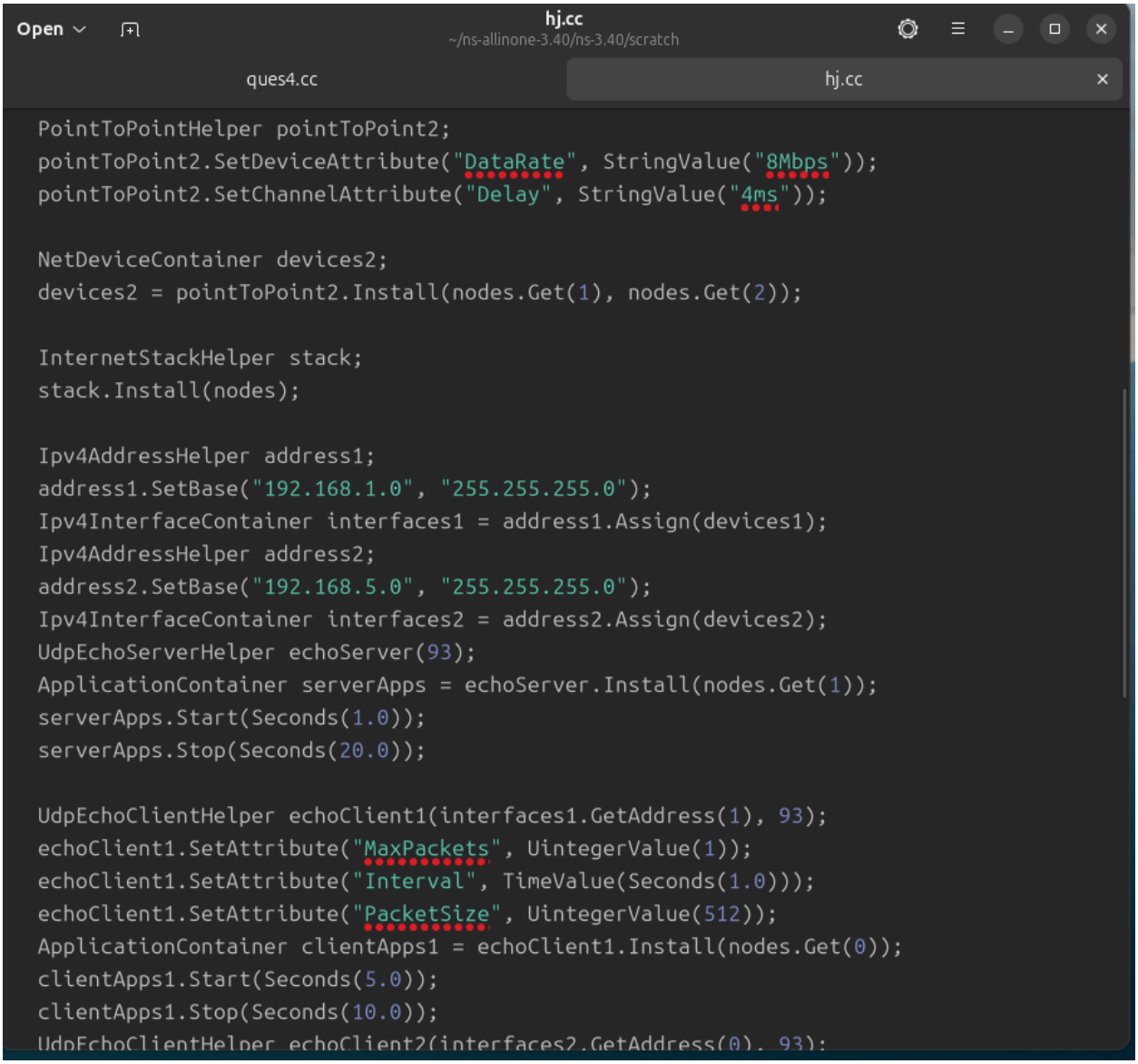
Time::SetResolution(Time::NS);

LogComponentEnable("UdpEchoClientApplication", LOG_LEVEL_INFO);
LogComponentEnable("UdpEchoServerApplication", LOG_LEVEL_INFO);
NodeContainer nodes;
nodes.Create(3);


 PointToPointHelper pointToPoint1;
 pointToPoint1.SetDeviceAttribute("DataRate", StringValue("8Mbps"));
 pointToPoint1.SetChannelAttribute("Delay", StringValue("8ms"));

 NetDeviceContainer devices1;
 devices1 = pointToPoint1.Install(nodes.Get(0), nodes.Get(1));


 PointToPointHelper pointToPoint2;
 pointToPoint2.SetDeviceAttribute("DataRate", StringValue("8Mbps"));
 pointToPoint2.SetChannelAttribute("Delay", StringValue("4ms"));
```


Open
hj.cc
~/ns-allinone-3.40/ns-3.40/scratch
ques4.cc
hj.cc
PointToPointHelper pointToPoint2;
pointToPoint2.SetDeviceAttribute("DataRate", StringValue("8Mbps"));
pointToPoint2.SetChannelAttribute("Delay", StringValue("4ms"));
NetDeviceContainer devices2;
devices2 = pointToPoint2.Install(nodes.Get(1), nodes.Get(2));
InternetStackHelper stack;
stack.Install(nodes);
Ipv4AddressHelper address1;
address1.SetBase("192.168.1.0", "255.255.255.0");
Ipv4InterfaceContainer interfaces1 = address1.Assign(devices1);
Ipv4AddressHelper address2;
address2.SetBase("192.168.5.0", "255.255.255.0");
Ipv4InterfaceContainer interfaces2 = address2.Assign(devices2);
UdpEchoServerHelper echoServer(93);
ApplicationContainer serverApps = echoServer.Install(nodes.Get(1));
serverApps.Start(Seconds(1.0));
serverApps.Stop(Seconds(20.0));
UdpEchoClientHelper echoClient1(interfaces1.GetAddress(1), 93);
echoClient1.SetAttribute("MaxPackets", UintegerValue(1));
echoClient1.SetAttribute("Interval", TimeValue(Seconds(1.0)));
echoClient1.SetAttribute("PacketSize", UintegerValue(512));
ApplicationContainer clientApps1 = echoClient1.Install(nodes.Get(0));
clientApps1.Start(Seconds(5.0));
clientApps1.Stop(Seconds(10.0));
UdpEchoClientHelper echoClient2(interfaces2.GetAddress(0), 93);

```
UdpEchoClientHelper echoClient1(interfaces1.GetAddress(1), 93);
echoClient1.SetAttribute("MaxPackets", UintegerValue(1));
echoClient1.SetAttribute("Interval", TimeValue(Seconds(1.0)));
echoClient1.SetAttribute("PacketSize", UintegerValue(512));
ApplicationContainer clientApps1 = echoClient1.Install(nodes.Get(0));
clientApps1.Start(Seconds(5.0));
clientApps1.Stop(Seconds(10.0));
UdpEchoClientHelper echoClient2(interfaces2.GetAddress(0), 93);
echoClient2.SetAttribute("MaxPackets", UintegerValue(1));
echoClient2.SetAttribute("Interval", TimeValue(Seconds(1.0)));
echoClient2.SetAttribute("PacketSize", UintegerValue(512));
ApplicationContainer clientApps2 = echoClient2.Install(nodes.Get(2));
clientApps2.Start(Seconds(11.0));
clientApps2.Stop(Seconds(15.0));
Simulator::Run();
Simulator::Destroy();
return 0;
}
```

```
ubuntu@ubuntu24-04:~/ns-allinone-3.40/ns-3.40$ ./ns3 run scratch/hj.cc
[0/2] Re-checking globbed directories...
ninja: no work to do.
At time +5s client sent 512 bytes to 192.168.1.2 port 93
At time +5.00854s server received 512 bytes from 192.168.1.1 port 49153
At time +5.00854s server sent 512 bytes to 192.168.1.1 port 49153
At time +5.01708s client received 512 bytes from 192.168.1.2 port 93
At time +11s client sent 512 bytes to 192.168.5.1 port 93
At time +11.0045s server received 512 bytes from 192.168.5.2 port 49153
At time +11.0045s server sent 512 bytes to 192.168.5.2 port 49153
At time +11.0091s client received 512 bytes from 192.168.5.1 port 93
ubuntu@ubuntu24-04:~/ns-allinone-3.40/ns-3.40$
```